Will AI replace us? How data scientists talk about AI taking their jobs

Dataset: Technological optimism or automation anxiety: how data scientists discuss AI-driven job displacement

LinkedIn posts by data scientists and AI practitioners discussing whether AI could replace their jobs, from Universitat Pompeu Fabra's CORA repository (doi.org/10.34810/DATA3307, CC0 license). It supports the 2026 Hipertext.net article "Technological optimism or automation anxiety: how data scientists discuss AI driven job displacement."

- 784 posts from April 2024 to June 2025
- 9 columns: raw text, cleaned text, and the probability of each post belonging to each of 7 LDA topics
- We named the topics from their top words: Will AI replace us?, LLM agents at work, AutoML automating ML, ChatGPT doing analysis, Vibe coding with GenAI, Juniors and citizen devs, Engineering skills for AI era

Data concerns: European CSV format (semicolons and decimal commas), 215 blank rows at the end of the file, 97 posts that lost emojis and fancy letters (saved as "????"), no dates, authors or sentiment labels, and 82% of posts in one topic.

Inspiration and Problem:

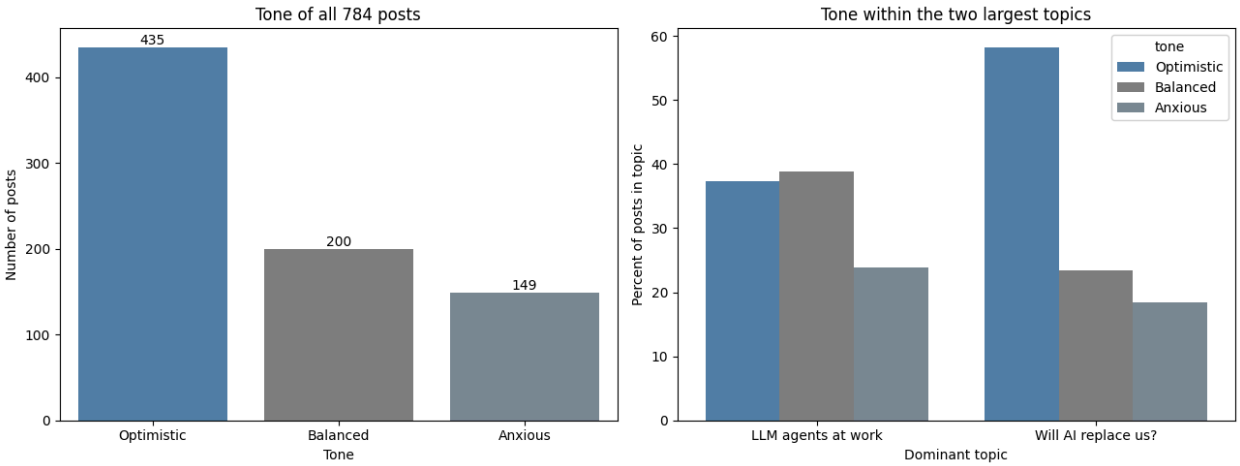
Fear of losing your job to AI is becoming a workplace health issue. About one in three U.S. workers expect AI to mean fewer job opportunities for them (Pew Research Center). From January to May 2026, U.S. employers blamed AI for 87,714 layoffs (Challenger, Gray & Christmas). Researchers at the University of Florida have proposed "AI Replacement Dysfunction" as a clinical framework for the stress caused by AI job insecurity.

Problem: there is no easy way to measure how worried workers are. We want to find out whether automation anxiety can be detected automatically in text, starting with data scientists, the people who build and understand AI best.

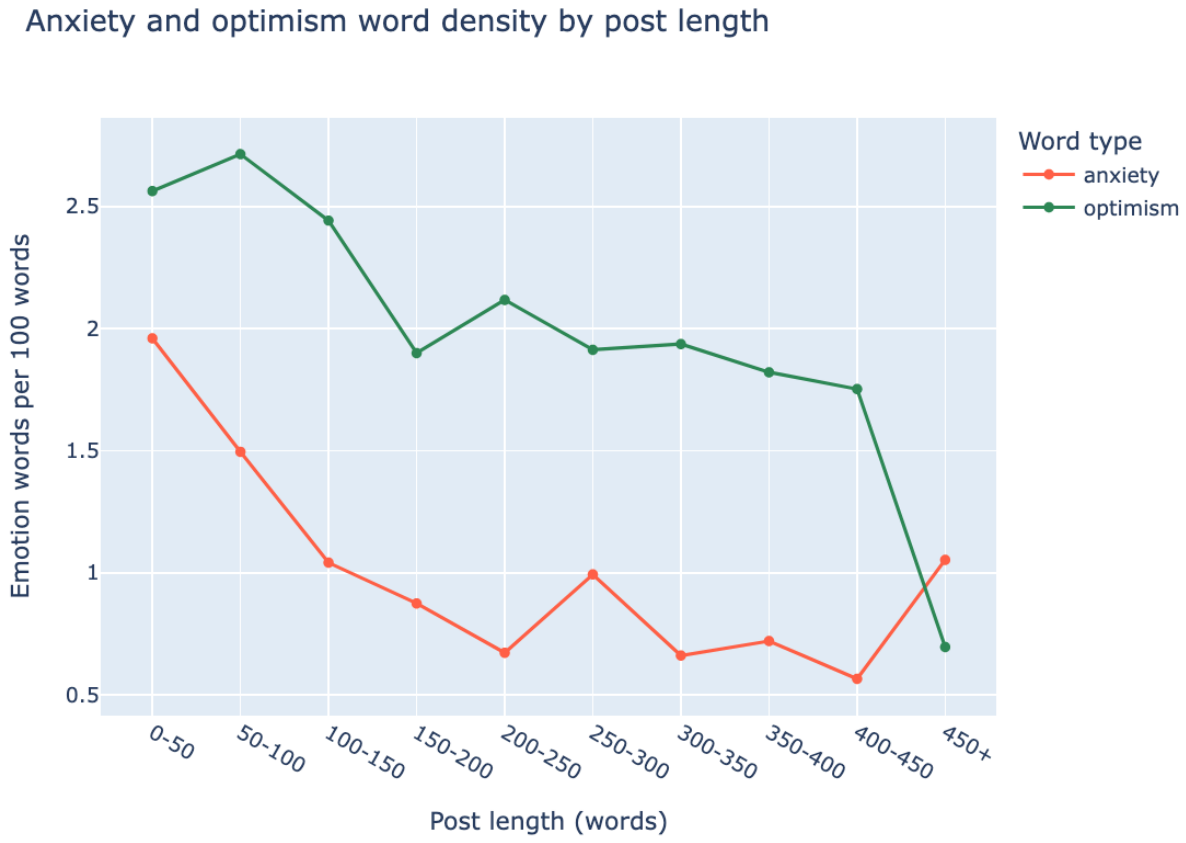
Exploratory Data Analysis:

1. Which words best separate anxious posts front optimistic ones?
    a. Compare the most frequent and most distinctive words (TF-IDF) in each group to see what people are actually afraid of.
2. Does a post's dominant topic predict its tone?
    a. Test whether the share of anxious posts really differs between topics
3. Do posts that mention specific roles (junior, analyst, engineer, scientist) sound more anxious than posts about the field in general?

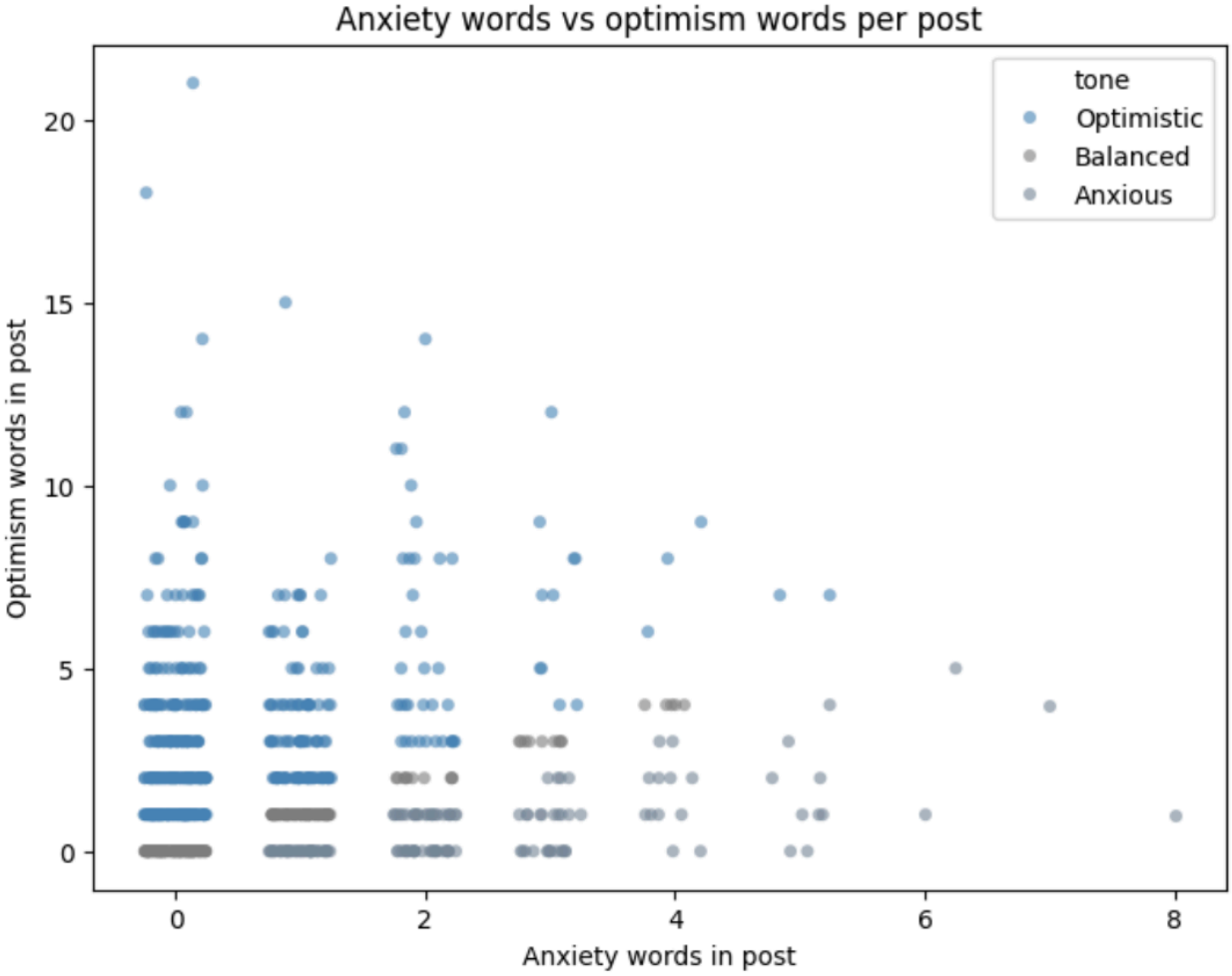
4. Do posts about learning and upskilling sound more optimistic than posts that never mention new skills?
    a. Test whether keep learning is how people cope with fear
5. How well does our simple word list agree with a real sentiment model?
    a. Label the posts with a zero shot transformer model and compare it with the word list labels to see where they disagree, especially on mixed posts.



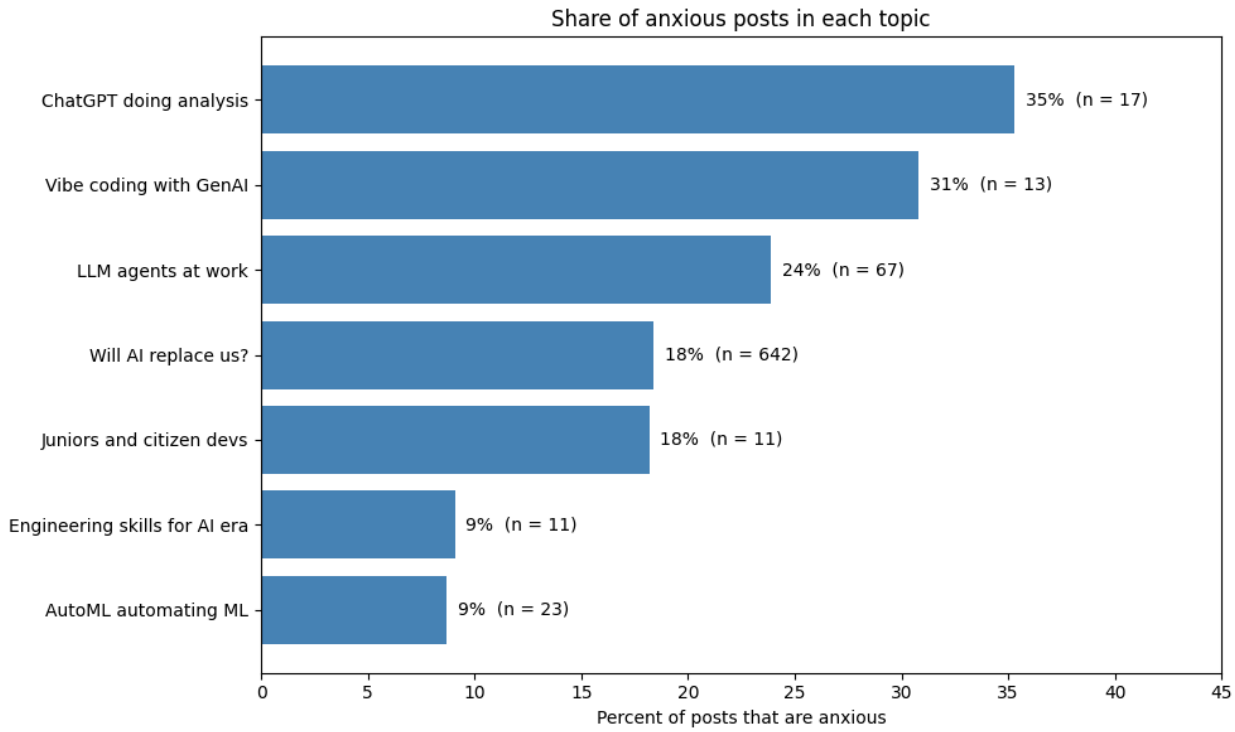
55% of posts lean optimistic, 26% are balanced, and 19% lean anxious. That matches the paper's finding that data scientists mostly see AI as a tool that helps their work.



Short posts use the most emotional language. As posts get longer and more technical, both worried and hopeful words drop off, but optimism stays higher at every length.



The two biggest topics move in opposite directions (correlation about −0.63), so a post usually sticks to one theme.



ChatGPT doing analysis (35% anxious) and vibe coding (31%) worry people most. AutoML and engineering skills are the calmest at about 9%. These groups only have 11 to 23 posts, though, so this needs a better model to confirm

Strategy

We will analyze 784 LinkedIn posts from the CORA / Dataverse CSUC dataset (DOI 10.34810/DATA3307, CC0 license), in which data scientists and AI practitioners discuss AI driven job displacement. Each post comes with its raw text, a cleaned version, and probability scores for 7 LDA topics.

- **Language and tools:** Python in Google Colab, with pandas for cleaning, seaborn and Plotly for visualization, and scikit-learn for modeling. Code and notebooks are shared on our team GitHub repository.
- **Algorithms:**
    - The LDA topic scores already come with the dataset.
    - We built a lexicon based tone label that counts anxiety words (fear, layoff, obsolete) against optimism words (opportunity, upskill, augment), giving each post an Optimistic, Balanced, or Anxious label.
    - We are deciding whether to upgrade this label with a zero shot or transformer sentiment model.
    - Logistic regression will be our main prediction model.
- **Prediction:** Can a post's topic profile (its 7 topic probabilities) plus its length predict its sentiment category?
- **What we hope to gain:** We want to find out whether AI and data science practitioners sound more anxious or more confident about AI taking their jobs, and whether that depends on which AI capability they are discussing, such as LLM agents, AutoML, or vibe coding.

Expected Results and Testing Strategy

Expected results:

- Topics will skew toward "Will AI replace us?" and "LLM agents at work." Our EDA already shows this: 82% of posts fall in the first topic and about 9% in the second.
- Sentiment will be mixed and ambivalent rather than uniformly negative. Our first pass labels show 55% optimistic, 26% balanced, and 19% anxious.
- Our model will give a modest but real improvement over a majority class baseline, not a strong classifier.

Testing strategy:

- We will hold out 20% of the posts as a test set (about 627 for training and 157 for testing) and keep the same tone proportions in both parts.
- On the training set, we will use k fold cross validation to divide the data into representative slices. Each model is trained and checked on every slice, so we can compare models fairly before we touch the test set.

- We will measure accuracy and F1 score against the majority class baseline. Always guessing "Optimistic" would be right about 55% of the time, so our model must beat that.
- We will recheck our findings after removing outlier posts that are very short or very long.
- We will manually spot check a sample of posts to see whether the tone labels match what the post actually says.

References

- Samokhina, O. (2026). Technological optimism or automation anxiety: how data scientists discuss AI driven job displacement. *Hipertext.net*, 32.
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- Blei, D., Ng, A., & Jordan, M. (2003). Latent Dirichlet Allocation. *Journal of Machine Learning Research*.
- Pew Research Center survey on U.S. workers and AI (the "1 in 3" figure).
- Challenger, Gray & Christmas (2026). Job cuts attributed to AI (the 87,000+ layoffs figure).
- Team GitHub repository: https://github.com/Nurislam1410/CS-133